

35-year-old pregnant lady at 32nd week of pregnancy presented for Caesarean section under general anaesthesia. She was diagnosed to have cerebellar arteriovenous malformation (AVM) at 16th week of pregnancy. At 28th week, there was bleeding from AVM rupture which was managed conservatively. She is planned for early endovascular embolization after the delivery.

A. How are you going to prepare this patient for the Caesarean section (30%) ?

B. Describe how you would conduct your anaesthesia (30%).

The surgery finished in one hour but patient failed to wake up.

C. What are the causes and how are you going to approach this problem (40%)?

A.

This is a pre-term, high risk parturient planned for LSCS under GA.

I will involve obstetrician, neurosurgeon/ interventional radiologist, and neonatologist in multi-disciplinary meeting.

To discuss optimal timing for LSCS

- Presumably 2x previous AVM bleeds (16th week and 28th week).
- Hormonal changes, BP swings, sympathetic surges during labour may increase risk of a rebleed.
- Need to balance fetal maturity/ growth with risk of rebleed if treatment is delayed.
- Arrange for LSCS to be done during office hours in elective list with experienced, dedicated anaesthetic manpower.

To discuss the definitive treatment of AVM post-LSCS

- NES and IR should be available during LSCS to standby for AVM rupture.
- Discuss with NES/ IR and patient whether angioembolisation should be performed in the same sitting as LSCS (post-delivery) as opposed to arranging for separate procedure.

Anaesthesia management plan

- RA (spinal) vs GA will need to be discussed with obstetrician, NES/ IR and patient.
- Assess for anticipated difficult airway to guide airway management.
- Assess spinal processes/ intervertebral space for CNB.
- Need for invasive lines (IA, possibly CVC).

Medical optimisation

- Baseline GCS, neurological status should be assessed and documented.
- Blood pressure control should be optimised to minimise shear stresses on AVM vessel walls.
- Sympathetic surges from pain, anxiety, stress should be avoided.
- Gestational complications (pre-eclampsia, GDM, placenta previa), if present, should be treated and optimised prior to surgery.
- Anaemia, if present, should be work up and corrected.
- Screen and optimise any other co-morbidities.
- Labs required include: FBC, RP, Coag panel
- GXM should be valid, with standby blood products.

Assessment of fetal well-being

- Risk of fetal abnormalities due to repeated exposure to X-rays during previous CT brain angiograms done as part of AVM work up.
- Uterus shields should have been used.
- Fetus should be screened for complications prior to LSCS.
- Need for steroids to aid lung maturation in pre-term baby (32nd week).
- Experienced neonatology consultant should be present during LSCS to attend to the baby post-delivery.

Patient, together with husband, will need to be counselled

- On anaesthesia options, need for invasive lines, post-op HD monitoring.
- High risk nature, with risks of potentially catastrophic bleed during LSCS
- Prioritization of patient care (mother's well-being will come before baby) in event of ruptured AVM.

B.

My anaesthesia of choice for LSCS is regional anaesthesia (spinal).

However, given the high risk of peri-op AVM rupture and plans for early angioembolisation post-delivery, I would opt for general anaesthesia with the following aims:

- Minimise haemodynamic swings to reduce risk of AVM rupture, including sympathetic surges from laryngoscopy, surgical incision, and during extubation.
- To minimise placental drug transfer from anaesthetic gases and opioids to pre-term fetus.
- Facilitate transition to angioembolisation post-LSCS if required.

Patient should be well fasted >8H, with prophylactic antacids (sodium citrate), and scheduled as an elective case during office hours. Baseline GCS and neurological status should be documented.

Experienced neonate team to be present in theatre to assess newborn, KIV need for neonatal HD/ ICU.

Monitors/ lines

- Large bore (at least 16G) IV cannulas in-situ, connected to hotline.
- Valid GXM, with standby blood products in OT.
- Pre-induction IA line, on top of standard ASA monitors (BP, HR, ECG, Fi/ ET O₂/ CO₂).
- Baseline ABG and ROTEM to be taken post-IA insertion.
- BIS to monitor depth of anaesthesia.
- Peripheral nerve stimulator to monitor TOFR.
- Post-induction temperature probe insertion.
- Discussion with obstetrician on use of cell saver if anticipated significant bleeding for LSCS.

Induction

- Patient positioned in slight left lateral tilt to relieve aorto-caval compression.
- Patient cleaned and draped, with obstetrician scrubbed prior to induction.
- Experienced neonatology consultant should be present in OT.
- NES and IR on standby.
- Airway topicalization with Lignocaine sprays to blunt laryngeal reflexes/ sympathetic surges.
- Pre-oxygenation with 100% O₂, targeting ETO₂ > 90%.
- RSI with Propofol 3mg/kg, Rocuronium 1.2 mg/kg, cricoid pressure, followed by insertion of appropriately sized cuffed ETT using video laryngoscope.
- Remifentanyl infusion to blunt sympathetic surges, with phenylephrine, Esmolol on standby to manage hyper/hypotension peri-induction.

Maintenance

- TCI propofol to minimise uterine relaxation (associated with inhalational agents).
- Remifentanyl infusion for surgical incision/ uterine manipulation until baby is out, followed by longer acting opioids (morphine), paracetamol, parecoxib (if Obs/ NES/ IR agreeable) after baby is delivered.
- Anti-emetics (dexamethasone, ondansetron) after baby is delivered.
- Passive and active warming with HME filter, warming blankets to avoid hypothermia.
- Adjust ventilation to avoid hyper/ hypocarbia, optimise FiO₂ to keep SpO₂ > 95%.

Bleeding/ Haemodynamic management

- TXA, uterotonics (syntocinon, carboprost) should be available and administered as appropriate.
- Avoid ergometrine due to its potential hypertensive effect.
- Monitor for bleeding, with ABG/ POCT Hb, ROTEM as required.
- Target SBP < 140 mmHg and minimise BP swings to lower risks of AVM rupture.
- Monitor for signs of AVM rupture (Cushing's reflex – hypertension with bradycardia, or sudden hypotension from blood loss) intra-op.
- IV fluids to ensure adequate preload.

Extubation

- Extubate awake after NMBA reversal guided by TOF monitor to achieve TOFR > 0.9 before extubation.

- Remifentanyl infusion can be kept to blunt sympathetic response until after extubation is done.
- Extended close monitoring in PACU for at least an hour.
- In particular, paying attention to haemodynamic stability, GCS, neurological status, seizures.
- Subsequent disposition to HD.

C.

Possible causes are:

Patient-related

Neuro:

- AVM rupture
- Stroke – ischaemic/ embolic/ haemorrhagic
- Seizure

Respi:

- Hypoxia from inadequate oxygenation
- Hypercarbia from inadequate ventilation

Metabolic:

- Hypoglycaemia
- Hyper/hyponatremia
- High urea
- Hypothyroid/ myxoedema coma

Hypothermia

Anaesthesia-related

- Residual anaesthesia agents
- Opioid overdose
- Inadequate NMBA reversal
- LA toxicity
- Magnesium toxicity

Obstetric-related

- Eclampsia
- Amniotic fluid embolism
- Post-partum haemorrhage
- Peri-partum cardiomyopathy

My approach would be:

This is a concerning situation.

Given the significant history of AVM, my top concern for failure to wake up would be AVM rupture.

If there are any signs of raised ICP (Cushing's reflex with hypertension and bradycardia, impaired pupillary response to light/ anisocoria), I would get surgeons to arrange for CT brain angiogram while I work up for other causes.

I would alert my senior, call for more help, and inform surgeons (OBS, NES, IR).

Airway

- Ensure ETT/ breathing circuit is patent.
- Check marking at lips, and auscultate for bilateral equal air entry to confirm proper placement.

Breathing

- Titrate FiO₂ to keep SpO₂ >95%
- Check for spontaneous breathing efforts.
- If not present/ inadequate, to optimise ventilation to keep ET CO₂ 35-40 cmh₂o

Circulation

- Check monitors to ensure BP, HR, ECG are normal.

Drugs

- Check and ensure TOFR > 0.9.
- High gas flows to wash out inhalational agents.
- Stop any propofol/ sedative/ opioid infusion.
- Review anaesthesia chart to check drugs given. Consider drug errors, drug interactions.
- Check for pinpoint pupils in opioid overdose. Given Naloxone as appropriate.

Apply BIS to check BIS values and EEG for signs of seizures.

- Monitor for any tonic-clonic movements, check for uprolling of eyes

Metabolic screen with POCT ABG – checking PaO₂, PaCO₂, Na, Glucose levels. Correct as appropriate.

Temperature probe to exclude hypothermia. Warm patient as required.

Subsequent management should be directed at cause for failure to wake.

Constant re-assessment until patient wakes up, or when CT scan room is ready to receive patient.